

The Scheme of the Wheelchair Dispatch and Cost Analysis for Epsilon Airlines

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Abstract

In this paper, we address the problem associated with the cost of airlines, such as the cost of wheel chairs, pay for the escorts and so on. We consider two models in hopes of minimizing the cost to the airlines. The cost includes the storage-cost for storing the wheel chairs in the concourses which depends on the using time and number of the wheelchairs, pay to escorts, maintenance cost of the wheelchairs and some potential cost.

Based on the available flight schedule downloaded from the Internet, the first model yields an approximate schedule of wheel chairs shifted from one concourse to another or from storehouse, which doesn't take the random characteristics of the problem into account and is basically a deterministic optimization model.

After presenting the first model, a more general, stochastic simulation model is established to analyze the shift process of the wheel chairs from one concourse to another, in which some variables such as the number of passengers who require help, the number of wheelchairs which need maintenance, and so on, are taken as stochastic ones. This model is more practical than the first model, and it is able to

adapt to different scaled aerodrome, so it has strong suitability, can be used to budget for short term.

Considering all potential costs and balance their respective weights, we create show third model. The model is more practical than the two models before. The result of simulation is so nice and feasible, it is the best model of the three models and fit for the budget for long term.

In fact, the above three models are able to handle a wide range of airports under a variety of circumstances. Also the minimum number of wheelchairs needed can be obtained to provide a good service and to minimize the total cost

Finally, we present a movement timetable of the wheel chairs, total cost analysis, passengers flow analysis and budget planning in short and long term.

Introduction

Most airports in the USA have wheelchairs available, and virtually all of the cities the airlines serve have jet bridges or special equipment to board customers who cannot climb stairs. Furthermore, airports have narrow aisle wheelchairs for customers who need to reach their seats in the airplane. When one passenger requests a wheelchair, airlines need to know if he is able to walk without assistance and how far,

and **whether** he can ascend and descend stairs without assistance.

Passengers can request a wheelchair by calling a phone number.

Restatement of the Problem

Airlines not only need to provide good service to passengers, but also want to minimize the service cost. So they provide wheel chair service to the passengers who require help. It is our goal in this paper to give out the movement timetable of the wheel chairs, total cost analysis and budget planning in short and long term.

During peak hours, which occurs typically when persons make their trip to and from work in larger cities, it is common that the number of the persons who require help will increase, so the airline should supply enough wheel chairs in time, avoiding flight delay. On the other side, the cost for supplying wheel chairs will increase.

We seek to balance these effects, along the cost with associated to offer extra wheel chairs, in order to provide reasonable recommendations for how to minimize the total cost by adjusting the number of wheel chairs.

Properties of a Successful Model

A successful schedule configuration should achieve the following problems:

- Minimize the total cost by adjusting the number of the wheel chairs

- Find a way to schedule the movement of wheel chairs through each day in a cost effective way.
- Determine all potential costs and balance their respective weights.
- Make a strong case as to why your solution is the best and show that it will be able to handle a wide range of airports under a variety of circumstances.
- Present the wheel chair shift status of the service station under high and low traffic loads.

General Assumptions

Let us formalize the assumptions which shall go into this model.

- There is a big storehouse, not far from the airport, used to store the wheel chairs which are not used at present. Because the chairs in the concourses are not in used will increase the total cost. And the cost for storing the chairs in the storehouse is so cheap that we consider it equal to zero.
- The cost for each wheel chair, transited from the storehouse to the airport has a linear relation with the transiting time, but the time spent on transiting wheelchairs is fixed.
- Once the persons who require help have been sent to the aircraft, the escorts return the nearest service station immediately.
- If the chairs are not enough, in order to reduce the cost for flight delay, the chairs must be sent to the airport several minutes ahead

of the flight landing time, as presented in the timetable, so that the airlines can offer wheel chairs quickly without delay.

- Without considering the traffic jams happens on the way to the airport.
- When some passengers will change to a different airplane, if the interval is short between the changing airplane taking on and the landing airplane taking off, you should serve these passengers first.
- Work efficiency does not distinguish between escorts.
- The number of the persons who require help does not exceed the total number of the wheel chairs, which includes in and out concourses.
- The cost for each chair shifted from storehouse to the airport is greater than the sum of the cost for shifting one chair from one concourse to another and storage-cost for each chair, so that the airlines try their best to shift chairs between concourses.
- The maintenance cost for each chair has linear relation with the using time.
- The cost for escorts shift chairs from one concourse to another has linear relation with the shifting time.
- The cost for flight delay has linear relation with the delay time.
- Several chairs, if needed, can be shifted to the concourse by the escorts simultaneously.

Cost optimization

Next, we seek a method of optimization that can be used to evaluate potential solution to the problem. How do we decide that the number of the wheel chairs will minimize the total cost? One natural way to compare potential solutions is to compute the total cost. It seems logical to conclude that well-designed model will require less money than the inefficient counterparts.

Suppose a model based on total cost minimization suggests that two hundred wheel chairs should be prepared for a large airport. Should operators heed this advice? Certainly, the operators of the airlines will incur significant cost in preparing and maintaining such wheel chairs. In crowded areas, it may not even be possible to find a place accept them. Furthermore, airlines employs escorts to serve the persons who require help without exact change. Paying for additional storage, maintenance, and labor costs may not be worth the added benefit of lowering the number of the wheel chairs.

This method must consider not only the number of the persons who require help , but also the agency operating the number of the wheel chairs.

We elect to use cost as a yardstick of our solutions. Cost is convenient medium due to its ubiquity in our culture. In considering the entire , we seek the facility configuration that will generate the lowest cost.

This cost will be distributed among both parties in our system—the airlines and the persons who require help.

In creating a cost optimization apparatus, we invoke the following terms and definitions:

- The general cost, $C[\text{dollars}]$, of an airlines is mainly consist of flight delay, denote $Q[\text{dollars}]$, the cost for shifting chairs from storehouse to the airport, denote $T[\text{dollars}]$, the cost for employ escorts, denote $S[\text{dollars}]$, the cost for chairs maintenance, denote $M[\text{dollars}]$, and the cost for shifting chairs out from airport to storehouse, denote $O[\text{dollars}]$.
- N is the number of the concourses in the airport.
- x_{i-j} ($i, j = 1, 2, \Lambda N$) is the cost for one escort shift a wheel chair from concourse i to concourse j in unit time.
- n_{i-j} ($i, j = 1, 2, \Lambda N$) is the number of the chairs, if needed, shifted from concourse i to concourse j . It is a variable.
- $flow_i'$ ($i = 1, 2, \Lambda N$) is the total number of the persons get out from aircraft to any concourse respectively.
- $flow_i$ ($i = 1, 2, \Lambda N$) [persons], is the number of the persons who require help who get out from the airplane to concourse i .
- $N(t)$ [chairs], is the number of the chairs, which are shifted from storehouse to the airport. (where t is the time of the flight schedule.)
- $N'(t)$ [chairs], is the number of the chairs, which are shifted from

airport to the storehouse.

- K [dollars/(chair×unit time)], is the cost for each chair shifted to the airport from the storehouse.
- K' [dollars/(chair×unit time)], is the cost for each chair shifted to the storehouse from the airport.
- m [dollars/(chair×unit time)], is the cost for chair storage.
- t_0 is the time of the first flight landing one day.
- t_l is the time of the last flight landing one day.
- qq is the cost for flight delay in unit time.

The underlying goal of this construct is to find a reasonable number of the wheel chairs, n_{i-j} that minimizes the total cost C .

Models

Model 1----- Deterministic Model

Assumption

- The ratio of the number of the persons who require help going to each concourse is equal.

Approach

We create the deterministic model in order to calculate the number of wheel chairs, which makes the minimized cost for the airlines, considering the storage-cost, cost for chairs transited from storehouse, and the cost for flight delay. One thing is certain that the number of the chairs is increasing, the cost for flight delay will

reduce. For another reason, the cost for flight delay is the greatest of all, so we construct the other function, $Q(\Delta t')$.

We now see that

$$C = S + T + Q + M + O$$

In order to make general cost minimized, it have to consider other cost, especially the shift cost(S), it is not a fixed scalar, so it is necessarily to create a model, considering all the cost.

$$\text{Min } C = S + T + Q + M + O$$

$$\text{st } \begin{cases} \sum_{i=1}^N n_{i-q} - \sum_{i=1}^N n_{q-i} + (N(t) - N'(t)) \geq \text{flow}_i & q=1,2,\Lambda, N \quad (*) \\ n_{i-q}, n_{q-i} \geq 0, \text{ and are integer variables} \end{cases}$$

where

$$S = \sum_{j=1}^N \sum_{i=1}^N x_{i-j} \times n_{i-j} \times t_{i-j} \quad T = \sum_{t=t_0}^t N(t) \times K \times t' + \sum_{t=t_0}^t N'(t) \times K' \times t''$$

$$M = \sum_{t=t_0}^l (B - N'(t) + N_0(t)) \times \Delta t \times m \quad N = 1,2,3,4,\Lambda \quad Q = qq \times (\Delta t')$$

expression (*) means that the number of the chairs must be greater than or equal to the number of the persons who require help, otherwise will increase the cost for flight delay. A simplified flow chart representing the above algorithm is presented in Figure 1

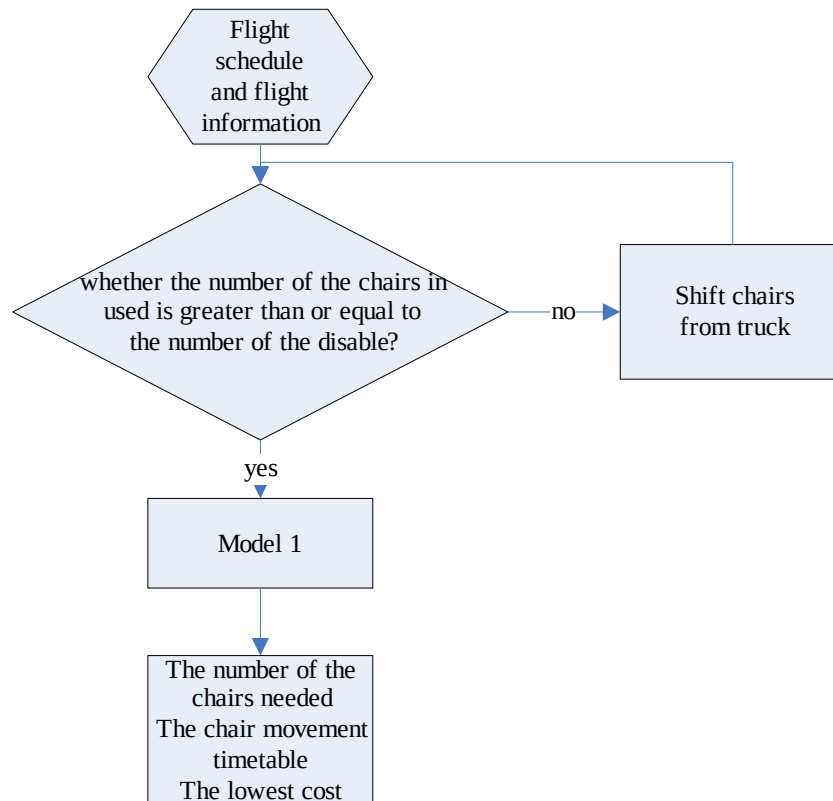


Figure1: arithmetic flow chart of model 1

Firstly, it is necessarily to acquire the number of the persons who require help from the flight schedule and the travelers' information, which has been known by the destination airport service station.

Next, the airlines will adjust the number of the wheel chairs by shifting chairs from storehouse or from concourses, if the total number of the chairs in used is less than the number of the persons who need help, it should transit chairs from storehouse until the total number of the wheel chair is greater than or equal to the number of the persons who require help, in order to reduce the cost for flight delay.

Finally, we' ll get the wheelchairs movement timetable, the number of the chairs need to be prepared by the airlines and the lowest cost.

Development of model

Random process has not been considered in this model, because we only wished for this model to serve as estimation value of the general cost. It is a deterministic model, a little rough, can be used to minimize the general cost approximately, but it has lowered the general cost much.

It should be noted that some scalars in this model, in reality, are not constant. For example, the $flow_i$ Q M will vary with time, they are random variables, are of certain distribution. In order to calculate accurately, we establish model 2 below.

Model 2----- Stochastic Model

Terms

- $flow_i'$ ($i=1,2,\Lambda N$) is the number of persons get out from flight to any , which is a random variable obey the Weibull distribution. Because there are one high traffic load, which happens at about 6:00PM, and low traffic load.
- $P_i(T)$ ($i=1,2,\Lambda N$) is ratio of the number of persons order the ticket with the number of seats in the aircraft. It is a random variable vary

the takeoff timetable.

- $Q_i(T)$ ($i=1,2,\Lambda N$) is the probability of the aircraft reach any apron, which obeys the uniform distribution.
- $R_i(T)$ ($i=1,2,\Lambda N$) IS ratio of the number of the persons who require help with $flow_i'$ in the same flight. It is also a random variable, varying with the takeoff timetable.
- $A_i(T)$ ($i=1,2,\Lambda N$) is the ratio of the number of persons who want to take another flight with $flow_i'$.
- $flow_i''$ is the number of the persons who will reach concourse i for another flight.
- $D_i(T)$ ($i=1,2,\Lambda N$) is the ratio of $flow_i''$ with the total number of the persons who need to change to a different airplane.
- $E_i(t)$ ($i=1,2,\Lambda N$) is the probability of the flight delay happens.
- $F_i(T)$ ($i=1,2,\Lambda N$) is the probability that the chairs need maintenance.

Optimization

Considering all parts of the cost and the random process, we present model 2 below:

$$\begin{aligned} \text{Min } C &= S + T + E_i(T) \times Q + M \times F_i(T) + O \\ \text{St } \begin{cases} \sum_{i=1}^N n_{i-q} - \sum_{i=1}^N n_{q-i} + N(t) - N'(t) \geq flow_i' \times A_i(T) \times R_i(T) \times D_i(T) \\ n_{i-q}, n_{q-i} \geq 0, \text{ and are integer variables} \end{cases} \end{aligned}$$

where

$$S = \sum_{j=1}^N \sum_{i=1}^N x_{i-j} \times n_{i-j} \times t_{i-j} \quad T = \sum_{t=t_0}^t N(t) \times K \times t' + \sum_{t=t_0}^t N'(t) \times K' \times t''$$

$$M = \sum_{t=t_0}^{tl} (B - N'(t) + N_0(t)) \times \Delta t \times m \quad q = 1, 2, \Lambda, N \quad Q = qq \times \Delta t'$$

A simplified flow chart representing the above algorithm is presented in Figure 2

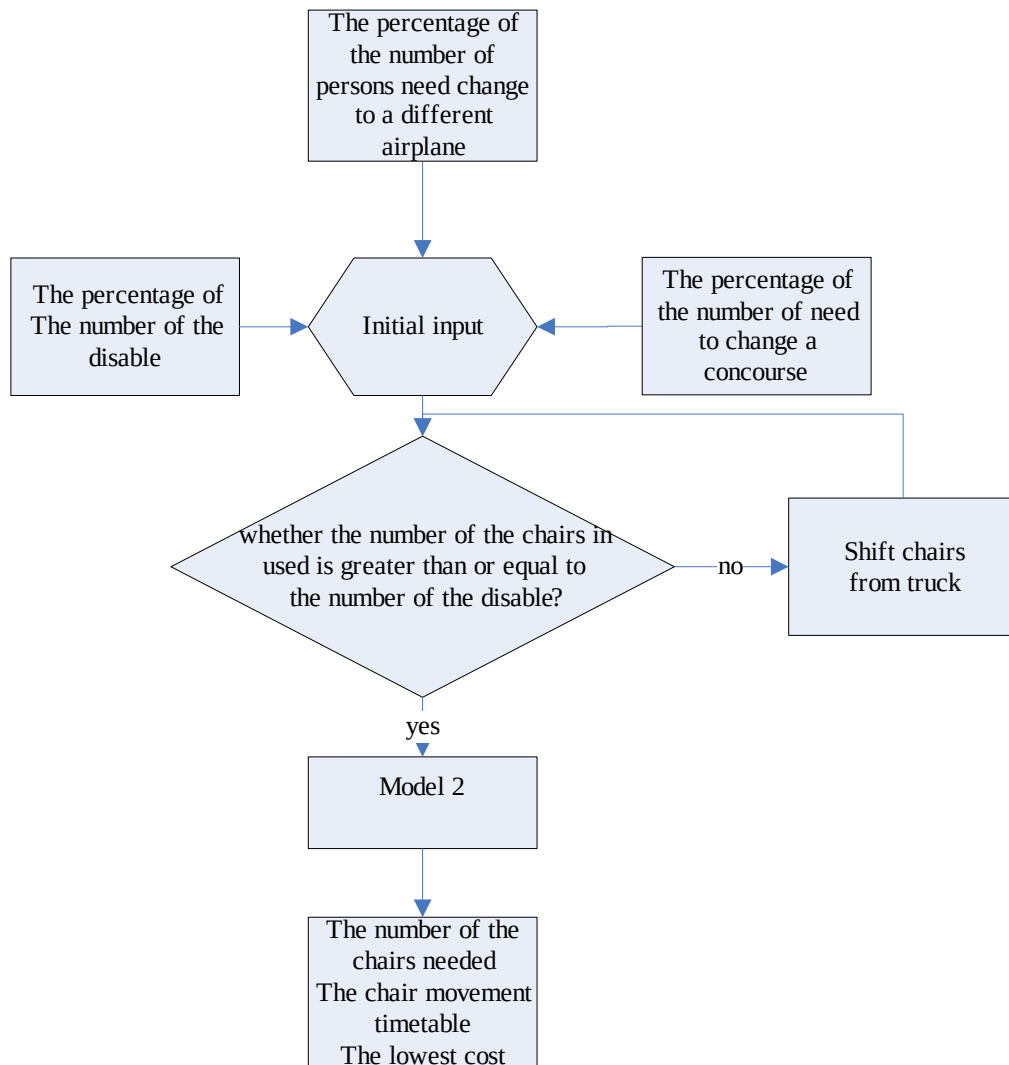


figure 2 arithmetic flow chart of model 2

Development of model

The random variables may not be of the distribution mentioned

above, however, they are considered as the function of the time, so these random variables can be instead of other variables, are of other distributions.

As it shown in the model 1 and model 2, $N = 1, 2, 3, 4, \Lambda$, which means the model is fit for different airports include large, medium and small airports.

The two models are not only fit for one airport, but also fit for several airports with several concourses, only if the airlines can offer enough wheelchairs.

Model 3---Time interval model

Considering all potential costs and balance their respective weights, we create the third model as follow:

$$\min\{x_{i-j}\} = \frac{C}{\Delta\zeta}$$

Where $\Delta\zeta$ the time span between the departure time and the time of a person need help was taken to the concourse.

Model discussion

Apparently, if the $\Delta\zeta$ equal to $\min\{x_{i-j}\}$ the general cost will be minimum. Because if $\Delta\zeta < \min\{x_{i-j}\}$ the escort can work slowly. To some degree, it is a waste in human resource. While $\Delta\zeta > \min\{x_{i-j}\}$ the cost for flight delay will increase. One hand is that the $\Delta\zeta$ is increasing, then the cost for flight delay will decrease, even the total cost will decrease. So we can say the more spent on the shifting-cost,

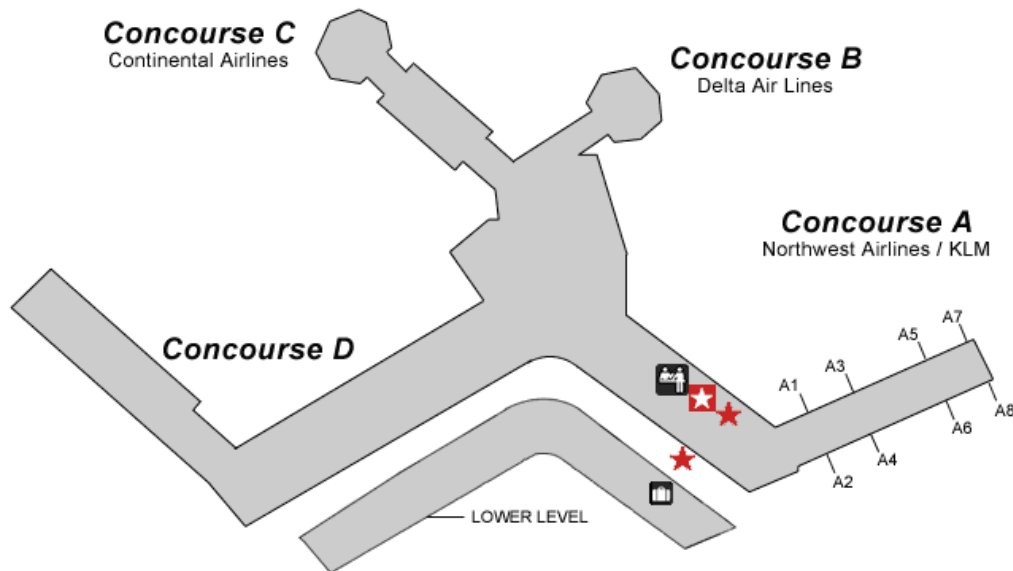
the lower general cost will be, in case that the number of the chairs is a little greater than or even equal to the number of the persons who need help.

Example and Simulation

As an example of the models , let us consider an airport with 4 concourses,

in the instance of Indiana.

Indianapolis International Airport (IND)



Airport Information	Notes
Indianapolis International Airport, 2500 South High School Rd, Indianapolis, IN 46241-4943 317-487-9594 www.indianapolisairport.com	International Arrivals can be picked up in either of the following locations: > International Arrivals Building (7001 Pierson Drive, Indianapolis, IN 46241) > Ground Transportation Center - Lower Level of the Main Terminal Tips to make your travel more enjoyable

Figure 3 Indianapolis International Airport

The simplify diagram for Indianapolis International Airport is shown in Figure4 below

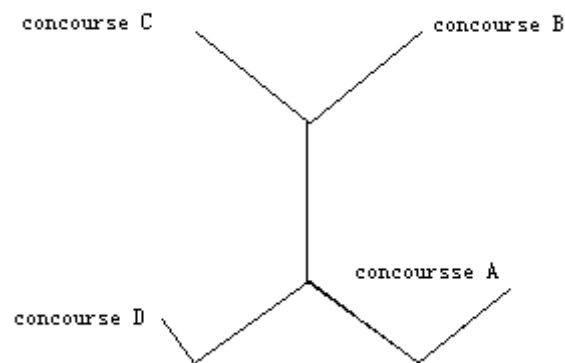


Figure 4 simplify diagram for Indianapolis International Airport

As it shown in the figure3, there are four concourses in the Indianapolis International Airport.

This is the flight schedule used to do simulation.

From: Dallas Metro TX US (DAL)
 To: Indianapolis IN US (IND)
 Date: 4 February 2006

Flight	From	Depart	To	Arrive	Via	Aircraft	Duration	inerary
DL561/DL962	DFW	5:30 AM	IND	11:15AM	ATL	M90/M80	4h45m	Add
CO214/CO288~	DFW	5:45 AM	IND	10:53AM	IAH	733/ERJ	4h08m	Add
NW410/NW1202	DFW	6:00 AM	IND	12:46PM	MSP	D9S	5h46m	Add
CO5853/CO5388*	DFW	6:00 AM	IND	12:46PM	MSP	D9S	5h46m	Add
DL1280/DL5197~	DFW	6:45 AM	IND	11:44AM	CVG	732/CRJ	3h59m	Add
DL766/DL4886~	DFW	7:00 AM	IND	1:36 PM	ATL	757/CRJ	5h36m	Add
DL6560/DL7092*	DFW	8:00 AM	IND	3:39 PM	MSP	319/D93	6h39m	Add
DL1664/NW3686*	DFW	8:05 AM	IND	1:36 PM	ATL	763/CRJ	4h31m	Add
CO314/CO2544~	DFW	8:15 AM	IND	1:49 PM	IAH	733/ERJ	4h34m	Add
NW6598/NW5144*	DFW	8:15 AM	IND	1:49 PM	IAH	733/ERJ	4h34m	Add
NW7228/NW5144*	DAL	8:30 AM	IND	1:49 PM	IAH	ERJ	4h19m	Add
CO2548/CO2544~	DAL	8:30 AM	IND	1:49 PM	IAH	ERJ	4h19m	Add
NW664/NW539	DFW	9:13 AM	IND	2:45 PM	DTW	319/D9S	4h32m	Add
CO5664/CO5311*	DFW	9:13 AM	IND	2:45 PM	DTW	319/D9S	4h32m	Add
DL6565/DL6950*	DFW	9:13 AM	IND	4:25 PM	DTW	319/D93	6h12m	Add
CO3265/CO2254~	DFW	10:45AM	IND	4:23 PM	IAH	ERJ	4h38m	Add
DL1056/NW9576*	DFW	11:00AM	IND	4:35 PM	ATL	763/CRJ	4h35m	Add
CO2029/CO2254~	DAL	11:00AM	IND	4:23 PM	IAH	ERJ	4h23m	Add
NW5109/NW5538*	DAL	11:00AM	IND	4:23 PM	IAH	ERJ	4h23m	Add
AZ3377/DL4276~	DFW	11:00AM	IND	4:35 PM	ATL	763/CRJ	4h35m	Add
NW638/NW468	DFW	12:23PM	IND	6:13 PM	DTW	D9S/D9S	4h50m	Add
DL6905/DL7320*	DFW	12:23PM	IND	6:13 PM	DTW	D93/D9S	4h50m	Add
CO5698/CO5468*	DFW	12:23PM	IND	6:13 PM	DTW	D9S/D9S	4h50m	Add
CO3191/CO2648~	DFW	12:30PM	IND	7:35 PM	IAH	ERJ	6h05m	Add
DL6420/NW825	DFW	2:35 PM	IND	9:58 PM	MCO	ERJ/D9S	6h23m	Add
NW3415/NW3794*	DFW	2:40 PM	IND	7:27 PM	CVG	CR7/ERJ	3h47m	Add
DL5015/DL6416~	DFW	2:40 PM	IND	7:27 PM	CVG	CR7/ERJ	3h47m	Add
CO2406/CO500	DAL	4:00 PM	IND	9:35 PM	IAH	ERJ/735	4h35m	Add
NW5480/NW5500*	DAL	4:00 PM	IND	9:35 PM	IAH	ERJ/735	4h35m	Add
NW3444/NW1859	DFW	4:20 PM	IND	8:50 PM	MEM	ARJ/320	3h30m	Add
DL6634/DL6555*	DFW	4:20 PM	IND	8:50 PM	MEM	ARJ/320	3h30m	Add
CO7385/CO6859*	DFW	4:20 PM	IND	8:50 PM	MEM	ARJ/320	3h30m	Add
DL559/DL507	DFW	5:10 PM	IND	11:08PM	ATL	M80	4h58m	Add
CO5807/CO5333*/NW4777~	DFW	6:34 AM	IND	2:20 PM	MEM-MCI	D9S/CRJ	6h46m	Add
DL2894/DL5707~DL5068~	DAL	8:30 AM	IND	4:03 PM	IAH-CVG	ERJ/CRJ	6h33m	Add
DL2960/DL5049~DL6416~	DAL	11:00AM	IND	7:27 PM	IAH-CVG	ERJ/CR7/ERJ	7h27m	Add

Figure 5 flight schedule

Passengers flow analysis

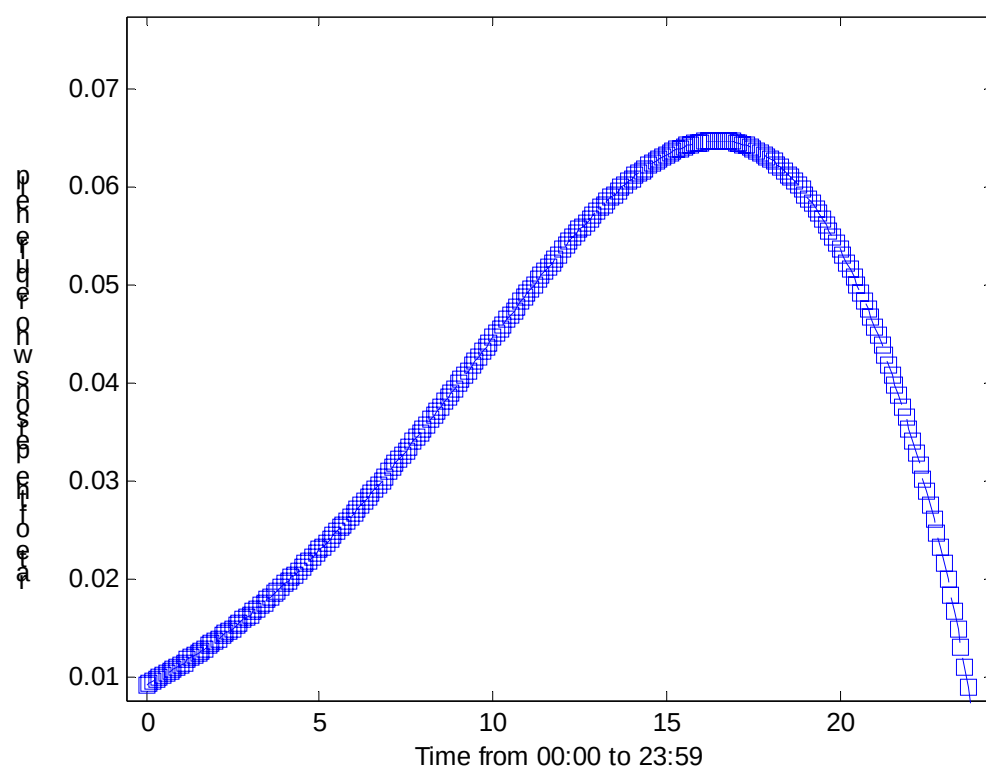


Figure 6 rate of the persons who require help---flight landing time

According to the flight schedule, we transform the data into the form of

the Figure 6. As it shown in figure 6, the rate of the persons who require help to the total person is distributed with the time. The rate will reach peak value at 6:13PM. Because, generally speaking, it takes about 6 hours to fly from Dallas to Indiana, the rush hour happens at about 10:00AM, it is convenient for the persons who need help to have a trip.

Simulation and results

Movement timetable

result of model 1

	10:53	11:15	11:44	12:46	13:36	13:49	14:20	14:45	15:39	16:03
a-a	0	0	0	0	0	0	0	0	0	0
a-b	0	0	0	0	0	0	0	0	0	0
a-c	0	0	0	0	0	0	0	0	0	0
a-d	0	0	0	0	0	0	0	0	0	0
a-e	0	0	0	0	0	0	0	0	0	0
b-a	0	0	0	0	0	0	0	0	0	0
b-b	0	0	0	0	0	0	0	0	0	0
b-c	0	0	0	0	0	0	0	0	0	0
b-d	0	0	0	0	0	0	0	0	0	0
b-e	0	0	0	0	0	0	0	0	0	0
c-a	0	0	0	0	0	0	0	0	0	0
c-b	0	0	0	0	0	4	0	6	0	0
c-c	0	0	0	0	0	0	0	0	0	0
c-d	0	0	0	0	0	21	0	2	0	0
c-e	0	0	0	0	0	0	0	0	0	0
d-a	0	0	0	0	0	3	0	2	0	0
d-b	0	0	0	0	0	0	0	0	0	0
d-c	0	0	0	0	0	0	0	0	0	0
d-d	0	0	0	0	0	0	0	0	0	0
d-e	0	0	0	0	0	0	0	0	0	0
e-a	0	0	0	0	0	16	0	0	0	0
e-b	0	0	0	0	0	16	0	0	0	0
e-c	0	0	0	0	21	17	0	0	0	0

d-a	0	0	0	0	0	0	0	5	0	0
d-b	0	0	0	0	0	0	0	0	0	0
d-c	0	0	0	0	0	0	6	0	1	0
d-d	0	0	0	0	0	0	24	19	18	18
d-e	0	0	0	0	0	0	0	0	0	0
e-a	0	0	0	0	0	22	0	13	0	0
e-b	0	0	0	0	0	27	0	8	0	0
e-c	0	0	0	0	21	0	0	0	0	0
e-d	0	0	0	0	0	9	0	0	0	9
e-e	0	0	0	0	0	0	0	0	0	0

Table 2.1 movement timetable of model 2

	16:23	16:25	16:35	18:13	19:27	19:35	20:50	21:35	21:58	23:08
a-a	42	42	0	0	30	20	16	16	0	0
a-b	0	0	18	0	0	0	0	0	0	0
a-c	0	0	0	0	0	0	0	0	0	0
a-d	0	0	27	0	0	10	4	0	0	0
a-e	0	0	0	0	0	0	0	0	36	0
b-a	0	3	0	0	0	0	0	0	0	0
b-b	28	25	25	43	43	43	43	43	11	0
b-c	6	0	0	0	0	0	0	0	0	0
b-d	0	0	0	0	0	0	0	0	0	0
b-e	0	0	0	0	0	0	0	0	32	11
c-a	0	0	0	0	0	0	0	0	0	0
c-b	0	0	0	0	0	0	0	0	0	0
c-c	0	5	5	42	42	40	40	40	0	0
c-d	0	34	0	0	0	2	0	0	0	0
c-e	0	0	0	0	0	0	0	0	40	0
d-a	0	0	0	30	0	0	0	20	0	0
d-b	0	0	0	0	0	0	0	0	0	0
d-c	33	0	22	0	0	0	0	0	0	0
d-d	0	0	12	9	9	9	21	5	0	0
d-e	0	0	0	0	0	0	0	0	5	0
e-a	0	0	0	0	0	0	0	0	0	0
e-b	0	0	0	0	0	0	0	0	0	0
e-c	0	0	15	0	0	0	0	0	0	0
e-d	0	0	0	0	0	0	0	0	0	0
e-e	0	0	0	0	0	0	0	0	0	0

Table 2.2 movement timetable of model 2

Result of model 3

Table 3.1[illegible]

e-a	0	7	0	0	0	0	0	0	0	0
e-b	0	0	5	0	0	0	0	0	0	0
e-c	0	0	30	0	0	0	0	0	0	0
e-d	0	0	0	0	0	0	0	0	0	0

Table3.2

The first column of the 6 tables means several chairs shifted from the former concourse to the latter concourse, for example, “c-a” means shift chairs from concourse C to concourse A. The number in the table means how many chairs must be shifted from former concourse to the latter concourse. The first row of the tables means the time of the flight landing. All the tables present how the escorts should move throughout each day.

Results Compare and Cost analysis

According the result above we get the compare table below:

	Model 1	Model 2	Model 3
Cost for escorts	12235\$	31100\$	34310\$

Table 4

As it shown in table 4 , the biggest cost for escorts is Model 3, that is to say the airlines operator spend more money on the cost for escorts to reduce the cost for flight delay, because the cost for flight delay is the biggest among the other cost .

We have discussed the general cost is consist of transit-cost, cost for shifting chairs, maintenance-cost and the cost for flight delay. Maybe

the deterministic model is not very precision, but in the long run, every random variables will tend to be stable. So we can use the model 1 for budget planning in a long term. Model 2 is much better precise, complete and suitability so it is fit for short term budget. Model 3 is the best model, for its precision, stability and ease to implement, so it is fit for budget for long term.

Discussion of models

Traffic load analysis

As it shown in Figure 6, the busy time is about between 4:00PM and 8:00PM.

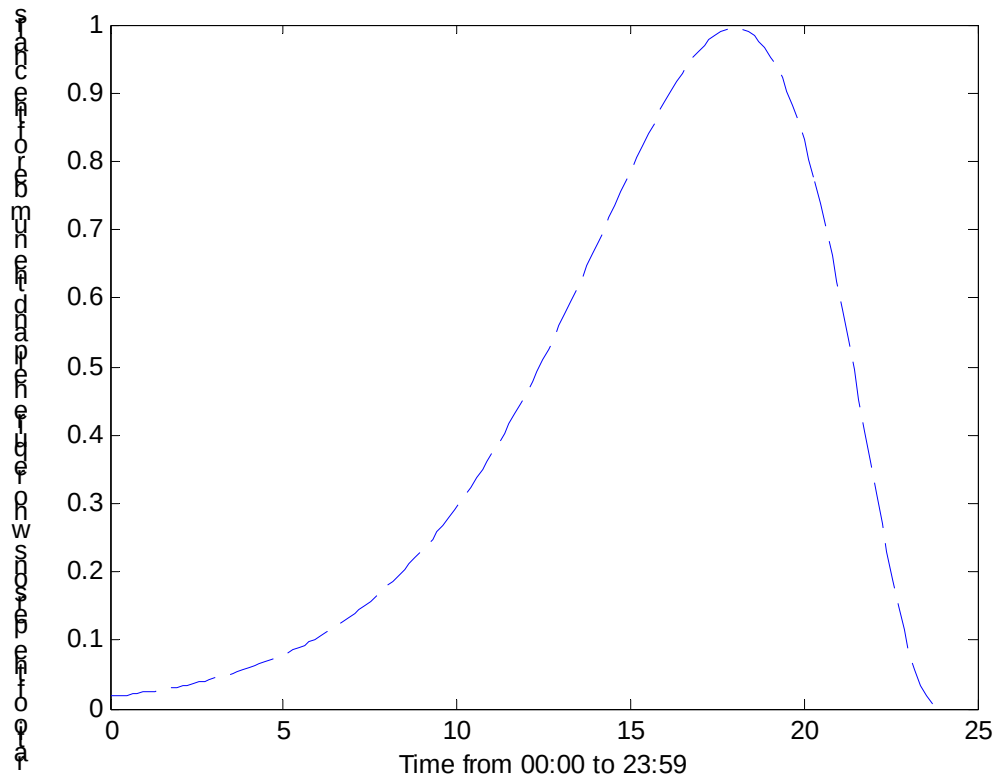


Figure 7 The ratio of the number of the persons who require help and the number of the chairs which must be prepared is distributed with the time.

Apparently, before 6:00PM, the airlines must transit wheelchairs to the airport time and time, otherwise, it will increase the cost for flight delay. After 6:13PM, chairs should be evacuated from the airport, otherwise it will increase the cost for storage-cost in concourses, which is pretty expensive.

Suggestions to Epsilon airlines

- Purchasing the wheel chairs which can be folded. Because it save more places to store more chairs.
- Hanging a big screen on the wall, so that the passengers in the concourse can read time. At the same time, it can display the flight schedule and the phone number of the service station.
- Equip several wireless phone in the corner of the concourse for the persons who require help who need help.
- If possible, blaze a green way for the persons who are a little late for the flight or the persons who require help.
- Finally, we suggest storing the wheelchairs not in used in the storage.

Conclusion

We used three models---deterministic model, accurate model and time

interval model--- in order to determine how to shift chairs at any time and the number of the chairs must be prepared, in order to cut down the general cost for Epsilon airlines.

The deterministic Model uses the simple schemes, the dynamic transition to transit chairs from storehouse or storehouse, ignoring random process, however provides us with an deterministic value, the least number of the chairs must be prepared for any given N concourses airport. The deterministic model looks at the motion of flight timetable as a whole, rather than individual models. It also finds rush hour by simulating flight landing time, above which the case of lacking the wheelchair will occur. Model 2 is a much more accurate model than the deterministic model, but the result of the simulation suggests that the general cost of the model 2 is a little higher than model 3.

In short, the methods to make the general cost minimized are as follow:

- Investigate the data include the number of the persons who require help in rush hour, to prepare enough chairs.
- Try best to shift chairs between concourses, if the number of the chairs is greater than or equal to the number of the persons who require help.
- Try to store the chairs to the storehouse.
- Adjust the work efficiency is another good way to reduce the general cost.

Potential Extension and Further Consideration

- As we have said that the models are not only fit in one airport, but also fit for several airports with several concourses.
- Considering the higher percentage of the old, the probability $R_i(T)$ will increase. Because the model 2 is so complete that this strong case will be disposed easily.
- Training cost for the escorts is one factor to the general cost.
- Traffic delay is a possible factor for the transit cost, even will increase the cost for the flight delay.
- The interval between the changing airplane taking on and the landing airplane taking off, is an important factor for the probability of flight delay. If the interval is small, the probability will increase, even increase the general cost. Oppositely, if the interval is large, the probability will decrease.

Strengths and Weakness

Strengths

- The results of our model are reasonable and the model is operational.
- Our model isn't sensitive to different initial values.
- Our model is suitable in extensive range, and it can be put into

implementation with different energies.

- Finally, our model fully considers the influences by the various factors, especially we have considered several random variables, so it can be used in some strong cases. These make our scheme more acceptable.

Weaknesses

- To simply the problem, there are some details we ignore.
- An obvious shortcoming of our model is that the parameters deterministic totally depend on the datum provided from different Internet addresses.

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